Nonlinear hydropower scheduling with coupled hydraulics and delayed river networks

Abstract

Short-term hydropower scheduling couples discrete operation with the movement of water and the conversion of hydraulic energy. We give a mixed-integer nonlinear formulation with finite hydraulic domains for reservoirs, shared pressurized waterways, generating units, discrete gates, instantaneous channels and delayed river networks. Storage evolves by an implicit midpoint balance. Pressure losses and efficiency-dependent generation remain nonlinear; supplied response curves have explicit interpolation semantics. Deterministic river transport preserves release pulses through confluences, whereas distributed transport uses conservative interval mixing. The solution method combines mixed-integer linear proposals, nonlinear dispatch, exact algebraic interpolation graphs and spatial branch-and-bound. Independent chronological integration separates physical acceptance from the objective-gap certificate of the finite optimization model. The formulation is implemented in Julia using open-source optimization libraries.

1 The scheduling problem

A generating schedule must account for water volume and hydraulic head. Discharging a reservoir removes water, changes its level, and changes the energy obtained from later discharges. Units also have operating ranges: an idle unit produces nothing, while a running unit cannot usually sustain arbitrarily small output. Head-dependent generation and these discrete operating domains have a long history in nonlinear hydropower scheduling [1].

We consider a deterministic horizon with given prices, inflows, equipment curves and operating restrictions. The physical model is deliberately quasi-steady. Pressure and discharge equilibrate within each control interval; reservoir storage evolves between intervals. Tunnel inertia, water hammer, free-surface river momentum equations, pumping, reserve products and stochastic decisions are outside its scope. River geometry is represented by supplied response functions, rather than resolved during scheduling.

The aim is a single explicit physical optimization problem. Linear approximations may help choose initial discrete decisions, but they do not define the delivered schedule's hydraulic feasibility. The final equations retain head-dependent generation, signed pressure losses and contemporaneous flow-dependent transport. Exactness below means equivalence to these declared functions and this time discretization, not exact reproduction of a real watercourse.

2 Volumes, heads and connections

Let $0 = \tau_0 < \cdots < \tau_T$ be time edges in hours, $\Delta_t = \tau_{t+1} - \tau_t$, and $m_t = (\tau_t + \tau_{t+1})/2$, for $t = 0, \ldots, T-1$. Operating inputs are piecewise constant, with changes on control edges. Use the sets $\mathcal{R}$ of reservoirs, $\mathcal{J}$ of pressure junctions, $\mathcal{B}$ of prescribed-head boundaries, $\mathcal{E}$ of tunnels, $\mathcal{C}$ of instantaneous channels, $\mathcal{G}$ of units and $\mathcal{D}$ of delayed reaches. Units belong to plants $\mathcal{G}_p$. Table 1 fixes the principal variables and units.

Head measures mechanical energy per unit weight. At a reservoir it is the water-surface elevation; at a flooded pressure junction it includes pressure head. Raising a parcel by one metre and losing one metre of head to friction have the same energy scale. Relative velocities and fixed local losses are absorbed into the supplied equipment relationships.

The unit conversions are

$$\kappa = \frac{3600}{10^6} = 0.0036, \qquad \gamma = \frac{\rho g}{10^6} = 0.00981 \quad (\rho = 1000 \text{ kg/m}^3,\ g = 9.81 \text{ m/s}^2). \tag{1}$$

Thus $\kappa \Delta q$ is a volume in Mm^3, and $\gamma q h$ is hydraulic power in MW.

Table 1: Physical variables. Storage and river inventory use time vertices; other indexed quantities use control intervals.

Variable	Meaning	Unit
V_{it}, W_{rt}	Reservoir volume, river water in transit	Mm^3
H_{nt}, h_{pt}	Node head, turbine net head	m
Q_{et}, X_{ct}, q_{gt}	Tunnel, channel, unit discharge	m^3/s
R_{rt}, d_{rt}	Reach entry flow, reservoir withdrawal	m^3/s
A_{rt}	Volume arriving from a delayed reach	Mm^3
P_{gt}	Electrical power	MW
u_{gt}, U_{pt}	Unit and plant operating states	$\{0,1\}$
b_{et}, a_{rt}	Tunnel position, reservoir outlet opening	$[0,1]$

Define F_{nt} as the net internal rate entering node n, before reservoir natural inflow. With incidence coefficients positive into a node,

$$F_{nt} = \sum_e B^E_{ne} Q_{et} + \sum_c B^C_{nc} X_{ct} + \sum_g B^G_{ng} q_{gt} + \sum_r D_{nr} \frac{A_{rt}}{\kappa \Delta_t} - \sum_r S_{nr} d_{rt}. \tag{2}$$

$D_{nr} = 1$ if reach r delivers into node n, and $S_{nr} = 1$ if a reservoir outlet at n supplies it. Other entries are zero. Tunnel and unit incidence uses the actual water destination: a river outfall withdraws from its source and injects the named reach, without immediately crediting the hydraulic target.

Reservoirs and pressure junctions obey, respectively,

$$V_{i,t+1} - V_{it} = \kappa \Delta_t (I_{it} + F_{it}), \qquad V_{i0} = V_i^{\mathrm{init}}, \tag{3}$$

$$F_{jt} = 0 \quad (j \in \mathcal{J}), \qquad \underline{V}_{it} \le V_{it} \le \overline{V}_{it} \quad (t = 0, \dots, T). \tag{4}$$

A junction cannot retain water, so its equation is a balance rather than an evolution law. Boundaries exchange water with the exterior and have fixed heads. Reservoir hydraulic heads use midpoint storage:

$$H_{it} = Z_i\left(\frac{V_{it} + V_{i,t+1}}{2}\right), \qquad \underline{H}_j \le H_{jt} \le \overline{H}_j, \qquad H_{bt} = H_b^{\mathrm{given}}. \tag{5}$$

Z_i is an increasing supplied level–volume table, or $Z_i(V) = z_i + s_i V + c_i V^2$. Midpoint storage is substituted into Z_i; averaging the two endpoint levels would give a different nonlinear model.

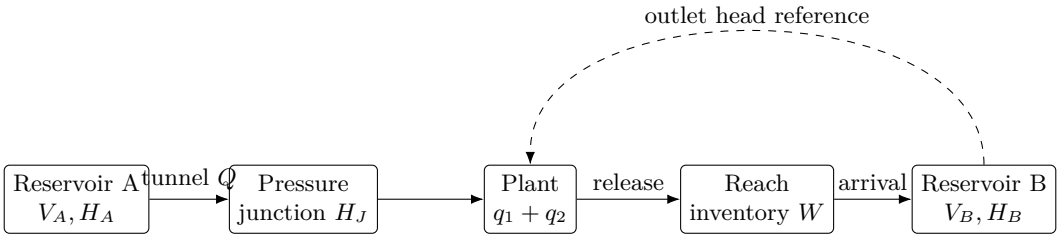


Figure 1: Water destination and hydraulic reference have different roles. Solid arrows move water; the dashed connection supplies the plant's outlet head. Water reaches B after transport, although its level already affects generation.

3 Pressure links and generating units

3.1 Tunnels and gates

For an oriented tunnel $e = (v, w)$, positive Q_e flows from v to w. At positive opening,

$$H_{vt} - H_{wt} = K_e(b_{et}) Q_{et} |Q_{et}|, \qquad K_e(b) = \begin{cases} R_e/b, & \text{without a gate-loss curve,} \\ R_e + G_e(b), & \text{with a supplied gate-loss curve.} \end{cases} \tag{6}$$

$R_e > 0$ and $G_e \geq 0$ have units $\mathrm{s}^2/\mathrm{m}^5$. The signed product ensures that friction opposes flow in either direction. Continuity and these pressure equations determine flow sharing through branches and loops; there is no independent allocation rule.

A closed gate instead imposes

$$b_{et} = 0 \implies Q_{et} = 0, \qquad |Q_{et}| \leq C_{et}. \tag{7}$$

It imposes no equality of endpoint heads. A directed river outfall also has $Q_{et} \geq 0$. Tunnel positions are either prescribed or selected from a finite declared set $\mathcal{S}_e \subset [0,1]$; arbitrary continuous tunnel positions are not optimization decisions.

An instantaneous channel between reservoirs has a different supplied law:

$$X_{ct} = \Phi_c(H_{vt} - H_{wt}, H_{\mathrm{ref},t}), \qquad |X_{ct}| \leq C_{ct}. \tag{8}$$

The reference is explicitly either endpoint level. Each constituent discharge curve is nondecreasing, passes through $(0,0)$ and preserves the sign of the level difference. Responses may be asymmetric. Interpolation is linear in level difference and between reference levels; endpoint segments continue linearly in difference, while the nearest reference curve is held outside the reference span. Channels have neither delay nor transit inventory.

3.2 Energy conversion

Let $q_{pt} = \sum_{g \in \mathcal{G}_p} q_{gt}$ and $P_{pt} = \sum_{g \in \mathcal{G}_p} P_{gt}$. All units at plant p share

$$h_{pt} = H_{\mathrm{source}(p),t} - \max\{H_{\mathrm{target}(p),t}, h_p^{\mathrm{floor}}\} - T_p(q_{pt}). \tag{9}$$

Omit the floor when absent. $T_p \geq 0$ is an additional tailwater rise, indexed by total plant discharge; it is not the complete downstream elevation. Shared penstocks are separate tunnels, whose loss therefore depends on the combined flow through the penstock.

Unit operation is a disjunction,

$$u_{gt} \in \{0,1\}, \qquad \underline{q}_{gt} u_{gt} \leq q_{gt} \leq \overline{q}_{gt} u_{gt}, \qquad \underline{P}_{gt} u_{gt} \leq P_{gt} \leq \overline{P}_{gt} u_{gt}, \tag{10}$$

$$P_{gt} = \gamma q_{gt} h_{pt} \eta_g(q_{gt}, h_{pt})\, e_g(P_{gt}). \tag{11}$$

On, additionally enforce

$$\underline{h}_g \leq h_{pt} \leq \overline{h}_g, \qquad \eta_g^{\min} \leq \eta_g(q_{gt}, h_{pt}) \leq 1, \qquad q_g^{\mathrm{lo}}(h_{pt}) \leq q_{gt} \leq q_g^{\mathrm{hi}}(h_{pt}) \tag{12}$$

The head and efficiency bounds always apply to running units; the flow envelopes apply when supplied by a turbine table. Off, $q = P = 0$ and these operating inequalities are inactive; hydraulic heads remain in the network.

Turbine efficiency is either a table or the analytic relationship

$$\eta_g(q,h) = \eta_g^0 - \alpha_g \left(\frac{q - q_g^*}{q_g^*} \right)^2 - \beta_g \left(\frac{h - h_g^*}{h_g^*} \right)^2. \tag{13}$$

Tables are linear in head and linear or shape-preserving cubic in discharge. Electrical efficiency e_g is a separate piecewise-linear function of *electrical output*. Hence (11) is implicit in P. The accepted electrical curves make $P/e_g(P)$ strictly increasing, so there is at most one admissible scalar conversion. If no curve is supplied, $e_g = 1$. This distinguishes turbine shaft conversion from electrical losses.

Plant state is the logical OR of its units:

$$U_{pt} \geq u_{gt} \ (g \in \mathcal{G}_p), \qquad U_{pt} \leq \sum_{g \in \mathcal{G}_p} u_{gt}, \qquad U_{pt} \in \{0,1\}. \tag{14}$$

Enforce $P_{pt} \leq \overline{P}_{pt}$, $q_{pt} \leq \overline{q}_{pt}$ when supplied, and $P_{pt} \geq \underline{P}_{pt} U_{pt}$, $q_{pt} \geq \underline{q}_{pt} U_{pt}$.

4 Reservoir outlets and river transport

4.1 Entry laws and confluences

A reservoir-source reach has total entry $R_{rt} = n_{rt} + d_{rt}$, where n_{rt} is its natural inflow and d_{rt} is withdrawal from the reservoir. One chosen outlet law is

$$d_{rt} = \begin{cases} (C_{rt} - n_{rt})a_{rt}, & \text{controlled outlet,} \\ c_r a_{rt}\sqrt{\ell_{rt}}, & \text{orifice,} \\ c_r \ell_{rt}^{3/2}, & \text{uncontrolled weir,} \\ a_{rt} D_r(H_{\text{source}(r),t}), & \text{supplied discharge curve.} \end{cases} \tag{15}$$

Here $\ell = H_{\text{source}} - z_r^{\text{crest}} \geq 0$, or $\ell = [H_{\text{source}} - z_r^{\text{crest}}]_+$ if dry operation is allowed. An uncontrolled weir has $a = 1$, including when a discharge table supplies its law. Other openings satisfy their declared bounds in $[0, 1]$. Ordinary reaches obey $n_{rt} \leq R_{rt} \leq C_{rt}$. Subtracting $d = R - n$, rather than R, in (2) counts natural river inflow exactly once.

For a confluence-fed reach, define local injection

$$L_{rt} = n_{rt} + \sum_{g \to r} q_{gt} + \sum_{e \to r} Q_{et}, \qquad R_{rt} = L_{rt} + \sum_{s \to r} \frac{A_{st}}{\kappa \Delta_t}. \tag{16}$$

Any number of upstream reaches and outfalls may merge. A confluence has one outgoing reach; it does not decide how to split a stream. Delayed reaches form a directed acyclic graph. Hydraulic loops through tunnels and instantaneous channels remain admissible.

4.2 Cohort response functions

Travel time separates release from arrival. The response-function viewpoint is classical [2]; here its precise discretization is part of the model. Reference flows are nonnegative, strictly ordered and cover admissible entry flows when more than one distribution is supplied. Delay edges are nonnegative and strictly increasing. For reference flow ρ_{rj}, a supplied delay distribution is uniform on each bin $[\delta_{rj\ell}, \delta_{rj,\ell+1}]$ with probability $w_{rj\ell} \geq 0$, $\sum_\ell w_{rj\ell} = 1$. Its cumulative distribution is

$$F_{rj}(s) = \sum_\ell w_{rj\ell} \min\left\{1, \max\left\{0, \frac{s - \delta_{rj\ell}}{\delta_{rj,\ell+1} - \delta_{rj\ell}}\right\}\right\}. \tag{17}$$

With two or more reference flows, the mixture weights satisfy

$$\theta_{rkj} \geq 0, \qquad \sum_j \theta_{rkj} = 1, \qquad R_{rk} = \sum_j \rho_{rj}\theta_{rkj}, \qquad \theta_{rk} \in \text{SOS2}\,. \tag{18}$$

Thus only neighboring distributions are blended. With a single distribution, $\theta_{rk1} = 1$ and there is no coordinate reproduction equation: its reference flow does not fix the release. The selected mixture stays with that cohort. Operating capacity changes do not rescale calibration reference flows.

For a rectangular cohort released during $[\tau_k, \tau_{k+1})$, the fraction arriving during interval t at reference j is

$$B_{rtkj} = \frac{1}{\Delta_k} \int_{\tau_k}^{\tau_{k+1}} [F_{rj}(\tau_{t+1} - s) - F_{rj}(\tau_t - s)]\, \mathrm{d}s. \tag{19}$$

The arrival volume and instantaneous arrival rate are

$$A_{rt} = A_{rt}^{\text{hist}} + \kappa \sum_k \Delta_k R_{rk} \sum_j \theta_{rkj} B_{rtkj}, \tag{20}$$

$$f_r(s) = f_r^{\text{hist}}(s) + \sum_k R_{rk} \sum_j \theta_{rkj} \left[F_{rj}(s - \tau_k) - F_{rj}(s - \tau_{k+1})\right]. \tag{21}$$

The integrals are evaluated exactly for the declared uniform bins. Interpolating the *fraction* and then multiplying by released flow makes the transfer piecewise quadratic in R; interpolating already released volumes would define a different response.

If every delayed reach has deterministic delay $\delta_r \geq 0$, use event-shaped entry streams $\widetilde{R}_r(s)$ and

$$f_r(s) = \widetilde{R}_r(s - \delta_r), \qquad \widetilde{R}_r(s) = L_r(s) + \sum_{s' \to r} f_{s'}(s), \qquad A_{rt} = \kappa \int_{\tau_t}^{\tau_{t+1}} f_r(s)\,\mathrm{d}s \tag{22}$$

at confluences for $s \geq 0$. For $s < 0$, $\widetilde{R}_r(s)$ is the independently supplied entry history of that reach, not a backward evaluation of the confluence recursion. On $[\tau_t, \tau_{t+1})$, a reservoir-source stream is $\widetilde{R}_r(s) = n_{rt} + d_{rt}$, including natural inflow. Link event streams to interval variables by

$$R_{rt} = \frac{1}{\Delta_t} \int_{\tau_t}^{\tau_{t+1}} \widetilde{R}_r(s)\,\mathrm{d}s. \tag{23}$$

Original pulses are shifted through successive confluences; they are not replaced by interval averages. Zero and sub-interval delays therefore produce genuine same-interval coupling. Entry-capacity restrictions are also enforced on the event stream, including both sides of its jumps.

If any reach uses a distributed response, (16) instead mixes arrivals into new interval-average downstream cohorts. Equations (19)–(20) remain exact for each such cohort, but downstream timing is a grid approximation. Refining the mixing grid and comparing cumulative arrivals and inventory estimates this error. Mass conservation alone cannot detect an incorrectly spread pulse.

4.3 Inventory, history and environmental flows

Released water that has not arrived is still in the system:

$$W_{r,t+1} = W_{rt} + \kappa \Delta_t R_{rt} - A_{rt}, \qquad W_{r0} = W_r^{\mathrm{hist}}. \tag{24}$$

Historical entry cohorts supply initial inventory and future historical arrivals: the historical terms in (20) and (21) use the same formulas over supplied past intervals. For a historical cohort k on $[s_k, e_k)$, let F_k be its selected delay mixture. Then

$$W_r^{\mathrm{hist}} = \kappa \sum_{k \in \mathcal{H}_r} R_{rk} \int_{s_k}^{e_k} [1 - F_k(-s)]\,\mathrm{d}s. \tag{25}$$

Unspecified earlier entry is zero in the declared model. Each reach's history already includes its pre-horizon upstream contributions; regenerating those contributions would double-count water. Restart retains original executed cohorts and their release flows, as well as reservoir states and operating histories. An average arrival bucket is insufficient to reconstruct future arrival timing.

Conservation identity. Summing (3) and (24) cancels every internal transfer. Thus

$$\sum_i (V_{i,t+1} - V_{it}) + \sum_r (W_{r,t+1} - W_{rt}) = \kappa \Delta_t \left(\sum_i I_{it} + \sum_r n_{rt} + E_t \right), \tag{26}$$

where E_t is net boundary exchange, positive into the modeled system. For example, a plant outfall withdraws q from its source, adds q to reach entry, and later leaves the reach as A; only that arrival enters the receiving reservoir. Equation (26) follows by this same cancellation for each arc, including reverse instantaneous flow.

Environmental restrictions may apply to reach entry $R_{rt} \geq R_{rt}^{\min}$, window-average arrival, or instantaneous arrival:

$$\int_J f_r(s)\,\mathrm{d}s \geq \int_J f_r^{\min}(s)\,\mathrm{d}s, \qquad f_r(s) \geq f_r^{\min}(s). \tag{27}$$

Only the selected arrival policy applies. For prescribed rectangular cohorts, distributed arrivals are piecewise linear; deterministic arrivals are piecewise constant. Checking response and requirement breakpoints and the relevant one-sided limits enforces the pointwise requirement for that transport model. At the horizon endpoints use the interior limits.

A flow observation is an additional inequality, not a water injection:

$$\sum_{g \in \mathcal{G}_o} q_{gt} + \sum_{r \in \mathcal{D}_o \cup \mathcal{C}_o} R_{rt} + I_{ot} \geq M_{ot}. \tag{28}$$

For an instantaneous channel, its observed release is $R_{ct} = X_{ct}$. Named contributions must refer to water already represented in the network.

5 Chronological operating constraints

Define unit starts and stops relative to the supplied initial state:

$$u_{gt} - u_{g,t-1} = s^+_{gt} - s^-_{gt}, \qquad s^+_{gt} + s^-_{gt} \le 1, \qquad s^\pm_{gt} \in \{0,1\}. \tag{29}$$

For every start at τ_t, enforce $u_{gk} \ge s^+_{gt}$ whenever $0 \le \tau_k - \tau_t < L^{\text{up}}_g$; for a stop, enforce $u_{gk} \le 1 - s^-_{gt}$ whenever $0 \le \tau_k - \tau_t < L^{\text{down}}_g$. Initial state age fixes the remaining dwell interval. Apply the same constraints to aggregate plant state U when plant dwell is declared. Changing the running member does not restart an already operating plant. Dwell beyond the horizon is carried to a restart, rather than forced by an invented terminal state.

Availability, forced states and prescribed output are hard restrictions:

$$\underline{u}_{gt} \le u_{gt} \le \overline{u}_{gt}, \qquad P_{gt} = P^{\text{given}}_{gt}, \quad q_{gt} = q^{\text{given}}_{gt} \tag{30}$$

where supplied. Plant state also satisfies $\underline{U}_{pt} \le U_{pt} \le \overline{U}_{pt}$; plant output schedules constrain sums. Maintenance forces the relevant units off. A zero schedule implies off; a positive schedule implies on. All restrictions intersect, including initial dwell; no input silently overrides another.

For interval-average discharge or release x_t, rate limits give

$$-\int_{m_{t-1}}^{m_t} r^-_x(s)\,\mathrm{d}s \le x_t - x_{t-1} \le \int_{m_{t-1}}^{m_t} r^+_x(s)\,\mathrm{d}s. \tag{31}$$

Unit and plant power use the same bounds with additional allowed jumps $\sum_{g\in\mathcal{G}_x} \underline{P}_{gt} s^+_{gt}$ upward and $\sum_{g\in\mathcal{G}_x} \underline{P}_{g,t-1} s^-_{gt}$ downward. These allow the minimum running output at a transition, not unrestricted startup or shutdown. Discharge and river ramps have no jump exemption. The first comparison uses a supplied previous-window average and midpoint. If absent, only that first comparison is omitted.

Reservoir volume ramps use $V_{i,t+1} - V_{it}$ and level ramps use $Z_i(V_{i,t+1}) - Z_i(V_{it})$, with rate integrals over $[\tau_t, \tau_{t+1}]$. They constrain vertex changes, not continuous-time derivatives.

For a tunnel interlock $\mathcal{E}_\ell$, with $o_{et} = \mathbf{1}[b_{et} > 0]$, require

$$\underline{N}_{\ell t} \le \sum_{e\in\mathcal{E}_\ell} o_{et} \le \overline{N}_{\ell t}. \tag{32}$$

A positive position counts as open even when its flow is zero.

6 Economic objective

With price π_t per MWh, reservoir and reach values v_i, v_r per Mm^3, and nonnegative movement and release-shortfall costs, maximize

$$\begin{aligned}\mathcal{V} = &\sum_t \pi_t \Delta_t \sum_g P_{gt} - \sum_{g,t}\left(c^+_{gt} s^+_{gt} + c^-_{gt} s^-_{gt}\right) - \sum_{e,t} c^b_{et} |b_{et} - b_{e,t-1}| \\ &- \sum_{r,t} c^r_{rt} \xi_{rt} + \sum_i v_i (V_{iT} - V_{i0}) + \sum_r v_r (W_{rT} - W_{r0}).\end{aligned} \tag{33}$$

The first gate comparison uses its given initial position. Transition and movement costs are charged once, without a duration multiplier. A positive release penalty makes the entry minimum soft:

$$\xi_{rt} \ge 0, \qquad \xi_{rt} \ge \kappa \Delta_t (R^{\min}_{rt} - R_{rt}). \tag{34}$$

ξ is a volume in Mm^3; positive cost makes it the actual shortfall. Without a positive penalty, the entry minimum remains hard and $\xi = 0$. Arrival minima and aggregate observations remain hard.

Terminal transit value prevents a release near the horizon from losing all economic value merely because it has not yet arrived. These values are explicit input assumptions. Subtracting initial inventories removes a constant endowment from both reported value and its relative-gap denominator.

7 Finite algebraic representation

The constraints above define a nonconvex MINLP. Discharge changes immediate power, future head and sometimes travel response. Neither fixed commitment nor exact interpolation makes these couplings convex.

7.1 Exact interpolation graphs

For a piecewise-linear curve with knots (x_i, y_i), use

$$\lambda_i \geq 0, \qquad \sum_i \lambda_i = 1, \qquad x = \sum_i x_i \lambda_i, \qquad y = \sum_i y_i \lambda_i, \qquad \lambda \in \mathrm{SOS2}\,. \tag{35}$$

SOS2 means that at most two nonzero weights are allowed, and their indices must be adjacent. It is a disjunctive interpolation constraint [4]. On one segment the coordinate equation determines those two weights, so (35) gives the original curve value exactly.

Turbine tables use independent discharge weights λ_i and head weights μ_j. Let E_{ij} be knot efficiencies. A shape-preserving cubic discharge interpolant [3] has the representation

$$\begin{aligned} w_i &= \lambda_i \lambda_{i+1}, & r_i &= \lambda_i w_i, \\ \eta_j &= \sum_i E_{ij}\lambda_i + \sum_i \left[A_{ij} r_i + B_{ij}(w_i - r_i)\right], & \eta &= \sum_j \mu_j \eta_j. \end{aligned} \tag{36}$$

For segment width $\delta q_i = q_{i+1} - q_i$, secant $d_{ij} = (E_{i+1,j} - E_{ij})/\delta q_i$ and original endpoint slopes m_{ij}, the coefficients are

$$A_{ij} = \delta q_i (m_{ij} - d_{ij}), \qquad B_{ij} = \delta q_i (d_{ij} - m_{i+1,j}). \tag{37}$$

On an active pair, $\lambda_i + \lambda_{i+1} = 1$, so the correction basis is $\lambda_i^2 \lambda_{i+1}$ and $\lambda_i \lambda_{i+1}^2$, which spans the cubic corrections that vanish at both endpoints. Off the active pair these products vanish. For bilinear tables $A = B = 0$. Clipping or extending a domain does not refit the supplied slopes; outer discharge extensions use their declared secants. Head extrapolation, where explicitly allowed, uses the outer head secant. Off-state efficiency extensions are finite algebraic continuations, not additional running-domain restrictions.

Compatible units share head coordinates, and each cohort shares one flow coordinate across arrivals, pointwise requirements and terminal inventory. The same construction represents piecewise-bilinear channel responses. Auxiliary equalities in (36), the power equation and released transport products remain nonlinear. SOS2 has not turned the problem into a MILP. These graphs preserve the specified functions; valid inequalities and branching improve the relaxation used to search them.

7.2 Quadratic tunnel and gate graphs

Signed losses can be expressed with a direction variable z_{et}:

$$Q = Q^+ - Q^-, \quad 0 \leq Q^+ \leq Cz, \quad 0 \leq Q^- \leq C(1-z), \quad S = (Q^+)^2 - (Q^-)^2, \quad z \in \{0,1\}. \tag{38}$$

Then $S = Q|Q|$. Directed outfalls need only $S = Q^2$; known head ordering can fix a direction. Finite gate selectors give both b and its open state o as affine expressions. Define $d_o = o(H_v - H_w)$. For bounds $L \leq H_v - H_w \leq U$, its exact binary-product hull is

$$Lo \leq d_o \leq Uo, \qquad H_v - H_w - U(1-o) \leq d_o \leq H_v - H_w - L(1-o). \tag{39}$$

Together with $|Q| \leq Co$ and $d_o = K(b)S$, this retains (6)–(7). $K(b)$ is affine in the finite position selectors, with any finite coefficient assigned to the closed position since $S = 0$ there. Only quadratic product equalities are needed. Wet/dry outlet branches similarly represent $\ell = \max(H - z^{\mathrm{crest}}, 0)$; an auxiliary root $r \geq 0$, $r^2 = \ell$, represents the square root in (15).

7.3 Bounds that retain the physical equations

Spatial branch-and-bound needs enclosing relaxations. Finite equipment and storage domains, conservation and forward/backward inventory bounds supply domains without restricting them to a previously found schedule. Polynomial bounds include interior extrema, not only table knots.

Let $\overline{e}$ bound electrical efficiency and define the running-domain power majorant $p^{\mathrm{ub}}(q,h) = \gamma \overline{e}\, q h \eta(q,h)$, with nonnegative q, h, η. It is polynomial on each interpolation cell, even when the electrical conversion is implicit. If $p^{\mathrm{ub}}(q,h) \leq aq + bh + c$, with $b \geq 0$ and a full head lower bound h_*, then

$$P \leq aq + b(h - h_*) + (c + bh_*)u \tag{40}$$

is valid for both states: on it bounds physical power; off $P = q = 0$ and the right side is nonnegative. Bernstein polynomial enclosures can bound the residual over each original interpolation cell to establish c. A plane fitted only at samples does not establish this inequality.

A joint box envelope couples discharge, running head and efficiency. Use exact binary-product hulls for $h_{\mathrm{on}} = uh$ and $\eta_{\mathrm{on}} = u\eta$. Nonnegative corner weights z_c satisfy

$$\begin{gathered}\sum_c z_c = u, \quad \sum_c q_c z_c = q, \quad \sum_c h_c z_c = h_{\mathrm{on}}, \quad \sum_c \eta_c z_c = \eta_{\mathrm{on}}, \\ \gamma e_{\min} \sum_c q_c h_c \eta_c z_c \le P \le \gamma e_{\max} \sum_c q_c h_c \eta_c z_c.\end{gathered} \tag{41}$$

Up to eight corners describe a nonnegative running box, and $e_{\min}, e_{\max}$ bound electrical efficiency on its power interval. A physical on point has product corner weights reproducing $qh\eta$; off all weights vanish. The exact efficiency and power equalities remain.

Table-aware supports also share on-state head weights ζ_j:

$$0 \le \zeta_j \le \mu_j, \quad \sum_j \zeta_j = u, \quad \sum_j (h_j - h_1)\zeta_j = h_{\mathrm{on}} - h_1 u, \qquad P \le aq + \sum_j c_j \zeta_j. \tag{42}$$

Coefficients enclose the complete polynomial power majorant, including corrections between head nodes. On, $\zeta = \mu$; off, $\zeta = 0$ while the hydraulic head stays free. Analogous discharge-coordinate supports couple the same power law along the other axis. A bounded separator may add further valid supports. These constraints tighten relaxations; none replaces a physical equality or adds a new operating decision.

8 Solution, reconstruction and certificate

JuMP supplies the algebraic Julia interface [5]. SCIP solves the global nonconvex model by branch-and-cut with spatial branching, nonlinear relaxations and bound tightening [7]. A preliminary MILP proposes unit states and finite gate positions using approximate generation and hydraulics. With those discrete decisions fixed, Ipopt solves nonlinear dispatch [6]. Several proposals can be dispatched under a shared time allowance. Only proposals passing reconstruction and finer replay are submitted as full-model starts. The MILP objective is a surrogate, never an enclosing bound for (33). Fixed-discrete dispatch is a piecewise-smooth nonconvex NLP, and a stationary point is not a global optimum.

Physical reconstruction then holds unit discharges and openings fixed and advances the coupled equations chronologically. For each interval, a Newton solve includes the next storage, pressure-junction heads, tunnel flows and active outlet releases. Current arrivals remain in the solve when travel time overlaps the same interval. Unobservable pressure components receive one numerical reference while retaining every water balance; admissible common pressure shifts respect junction bounds. Electrical output is recovered from (11). Reusable numerical arrays separate this equation evaluation from input parsing and solver reporting.

Acceptance checks the original interval equations and operating restrictions, then repeats integration on a finer grid with the original controls. Schedules and ramps are compared over their original operating windows; storage vertices, total water, cumulative arrivals and terminal transit are checked separately. This prevents an apparently converged pressure solve from concealing an operating violation or a transport error. For prescribed tolerances $\varepsilon_V, \varepsilon_{\mathcal{V}}$, consistency also requires

$$\begin{gathered}\max_{i,t} |V_i^{\mathrm{fine}}(\tau_t) - V_{it}| \le \varepsilon_V, \qquad \max_r |W_r^{\mathrm{fine}}(\tau_T) - W_{rT}| \le \varepsilon_V, \\ |\mathcal{V}^{\mathrm{fine}} - \mathcal{V}| \le \varepsilon_{\mathcal{V}} \max(1, |\mathcal{V}|).\end{gathered} \tag{43}$$

Refinement supplies a numerical consistency test, not a rigorous enclosure of continuous-time hydraulic error.

Let L be the independently reconstructed value of a schedule feasible for the declared interval model and U a finite enclosing solver upper bound. The reported gaps are

$$g_{\mathrm{abs}} = \max(0, U - L), \qquad g_{\mathrm{rel}} = \frac{\max(0, U - L)}{\max(1, |L|)}. \tag{44}$$

A numerical certificate requires an enclosing bound and the requested absolute or relative gap. It concerns the declared interval model, interpolation and finite gate sets. Fixing discrete decisions makes it conditional on those decisions. The interval-model certificate and finer replay acceptance are separate outcomes: a schedule with a certified objective gap can still fail finer replay. Neither certifies a continuous-time optimum. Floating-point solver tolerances and guarded polynomial bounds do not constitute an interval-arithmetic proof.

9 Discussion

The useful separation is between the physical model, methods for finding feasible controls, and methods for enclosing their optimum. Direct nonlinear dispatch keeps head and efficiency changes inside the scheduling problem; cohort accounting preserves water and its timing; exact interpolation graphs retain the supplied response rather than silently fitting another one. Independent reconstruction gives the delivered controls their physical meaning, while global bounds quantify unresolved economic improvement.

These choices do not remove nonconvexity, make coarse river mixing exact, or guarantee a short proof on a large network. Their benefit is a model whose approximations and certificates can be stated together. Site calibration, unmodeled hydraulic regimes and uncertainty remain separate questions. The formulation can be extended by adding explicit physical laws and operating constraints while retaining the same conservation, interpolation and acceptance principles.

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